Technology Stack

Date	15 March 2026
Team ID	SWTID-2026-6686
Project Name	Streamlining IT Procurement: Automating Standard Laptop Orders with Flow Designer
Team Leader & Members	**Leader:** MANISHA R (Size: 4) **Members:** GOPIKA K, VINDHIYA G, NITHIYASRI M
Maximum Marks	2 Marks

Define Your Technology Stack:

Identify the programming languages, frameworks, databases, front-end tools, back-end tools, and APIs that you will use to build your solution. A well-chosen technology stack ensures that your application is scalable, maintainable, and aligned with your project requirements.

Fill out the table below to detail the specific technologies chosen for each component of your architecture and provide a brief justification for your choice.

Technology Stack Details

S.No	Architecture Component / Layer	Technology Chosen	Justification / Purpose
1	**Frontend / Client-Side** *(e.g., React, Angular, HTML/CSS)*	*ServiceNow Service Portal, HTML5, CSS3, AngularJS / Bootstrap*	Provides an intuitive, responsive, and customizable self-service catalog interface for employees to select and request standard laptop models seamlessly.
2	**Backend / Server-Side** *(e.g., Node.js, Python/Django, Java)*	*ServiceNow Flow Designer, JavaScript (Server-side Scripts)*	Enables low-code automation for orchestrating approval workflows, task assignment, order routing, and SLA tracking without complex custom coding.
3	**Database / Data Storage** *(e.g., MySQL, MongoDB, PostgreSQL)*	*ServiceNow System Tables (CMDB & Asset Management, PostgreSQL engine)*	Provides strong relational structure for managing asset configuration items, procurement catalog records, inventory stock levels, and user roles.

S.No	Architecture Component / Layer	Technology Chosen	Justification / Purpose
4	**Cloud / Hosting / Deployment** *(e.g., AWS, Azure, Heroku, Docker)*	*ServiceNow SaaS / PaaS Cloud Instance*	Ensures enterprise-grade security, high availability (99.9%), automated scaling, and zero setup overhead for hosting procurement workflows.
5	**Version Control & CI/CD** *(e.g., GitHub, GitLab, Jenkins)*	*GitHub & ServiceNow Update Sets / Studio*	Allows seamless multi-developer collaboration, versioning of Flow Designer workflows, and automated deployment across Dev, Test, and Production instances.
6	**Third-Party APIs / Other Tools** *(e.g., Stripe, Twilio, Google Maps)*	*REST / SOAP Webhooks, IntegrationHub, SMTP Email Gateway*	Enables real-time automated notifications to users and external hardware vendor fulfillment systems upon order approval.